

LearnHub: Your Center for Skill Enhancement

SHORT TERM INTERNSHIP PROGRAMME - By SmartBridge

Category : Full Stack Development(MERN)

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2. PROJECT OVERVIEW

LearnHub: Your Center for Skill Enhancement is a mobile and web-based e-learning platform designed to help users—especially from rural and underserved communities—gain essential digital skills, improve their employability, and enhance their quality of life through accessible, localized education.

• PURPOSE

The purpose of LearnHub is to empower individuals by providing a user-friendly, interactive platform that supports self-paced learning and bridges the digital literacy gap. It aims to simplify access to essential knowledge and skills in areas such as computer usage, internet safety, mobile applications, and communication.

- 1. Empowering Users with Knowledge:**
LearnHub offers structured, multilingual lessons with practical examples to help learners understand and apply digital concepts confidently in real life.
- 2. Simplifying the Learning Journey:**
Through step-by-step tutorials, visual aids, and voice-assisted instructions, LearnHub makes digital education easy for learners of all ages and literacy levels.
- 3. Supporting Skill Growth:**
LearnHub provides curated learning tracks in areas such as digital literacy, smartphone use, typing, online services (e.g., UPI, banking), and soft skills to prepare users for jobs or entrepreneurship.
- 4. Assisting Trainers and Educators:**
The platform serves as a teaching and monitoring tool for teachers, NGOs, and government volunteers, helping them guide learners and track progress efficiently.
- 5. Creating an Inclusive Learning Space:**
By offering offline access, low-bandwidth support, and regional language content, LearnHub ensures everyone—regardless of their location or background—can learn, grow, and thrive.

• FEATURES OF LEARNHUB:

LearnHub is equipped with various features designed to improve the learning experience, support trainers, and ensure measurable progress. These features help

learners engage with content, track their growth, and stay motivated throughout their educational journey.

1. Smart Course Navigation

- **Filtered Course Paths:** Learners can choose courses based on their goals—digital basics, mobile literacy, internet safety, etc.
 - **Saved Progress:** Learners can resume courses from where they left off and track their module-wise completion.
-

2. Interactive Learning Modules

- **Audio-Visual Content:** Lessons include videos, images, and voice explanations to enhance understanding.
 - **Gamified Quizzes:** Engaging assessments after each module to reinforce learning and boost confidence.
 - **Practice Tasks:** Real-life simulations like sending an email or using a QR code.
-

3. Localised and Inclusive Design

- **Multilingual Support:** Content is available in regional languages for better reach and comprehension.
 - **Voice Assistance:** Helpful for users with low literacy—navigation and lessons can be guided through audio.
 - **Offline Learning:** Downloadable lessons allow access without constant internet availability.
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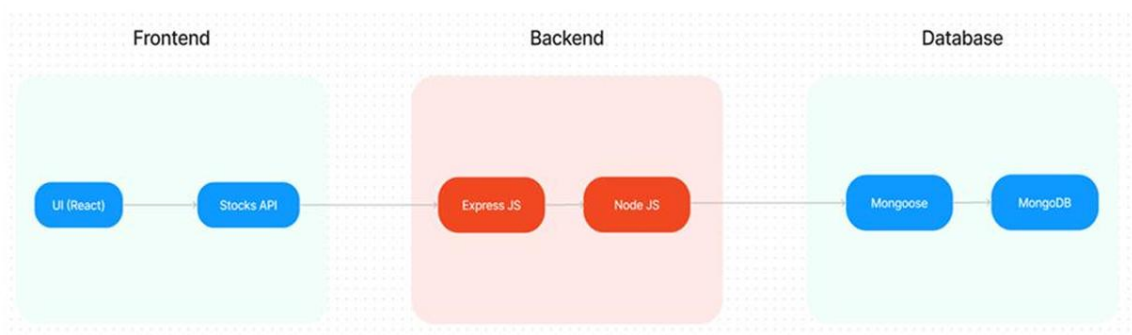
4. Progress Tracking and Certification

- **Digital Report Cards:** Learners can view their completed modules, scores, and milestones.
- **Completion Certificates:** Learners earn shareable digital certificates after finishing a course.
- **Leaderboard/Badges:** Optional gamification elements to motivate and recognize top learners.

5. Community and Support Tools

- **Trainer Dashboard:** Facilitators can enroll learners, monitor activity, and assess performance.
- **Doubt Forums & Help Chat:** Learners can post questions and receive help from trainers or peer learners.
- **Mentor Connection:** In advanced versions, learners can connect with mentors or guest experts.

3. TECHNICAL ARCHITECTURE



The technical architecture of LearnHub follows a client-server model, where the frontend acts as the client and communicates seamlessly with the backend server. This architecture ensures efficient delivery of educational content, real-time user interaction, and secure data management, creating an inclusive and scalable learning environment.

Frontend (Client-Side)

The frontend of LearnHub is built using React.js, providing a dynamic, component-based UI experience. Users interact with the platform through intuitive layouts designed with Material UI and Bootstrap, ensuring a responsive and accessible interface across all devices, especially mobile phones and low-end hardware common in rural areas.

To fetch learning content, track progress, and manage authentication, the frontend uses Axios to communicate with the backend via RESTful APIs. Learners, educators, and administrators each access customized interfaces based on their roles.

Backend (Server-Side)

The backend is powered by Node.js and Express.js, which handle routing, authentication, content delivery, progress tracking, and user management. The server processes API requests from the frontend, manages user sessions, and connects to the database layer to fetch or store learning data.

Security, scalability, and performance optimization are key focuses of the backend architecture, allowing LearnHub to serve both low-volume individual learners and high-volume institutional deployments.

Database (Data Layer)

The platform uses MongoDB, a flexible NoSQL database, for efficient storage of structured and unstructured educational data. This includes:

- User profiles (learners, trainers, admins)
- Learning modules and course content
- Quiz results and progress reports
- Certificates and achievements

Mongoose is used as an Object Data Modeling (ODM) library to simplify interaction with MongoDB. It ensures schema

enforcement, validation, and streamlined queries, making backend development faster and more robust.

Key Technologies Used

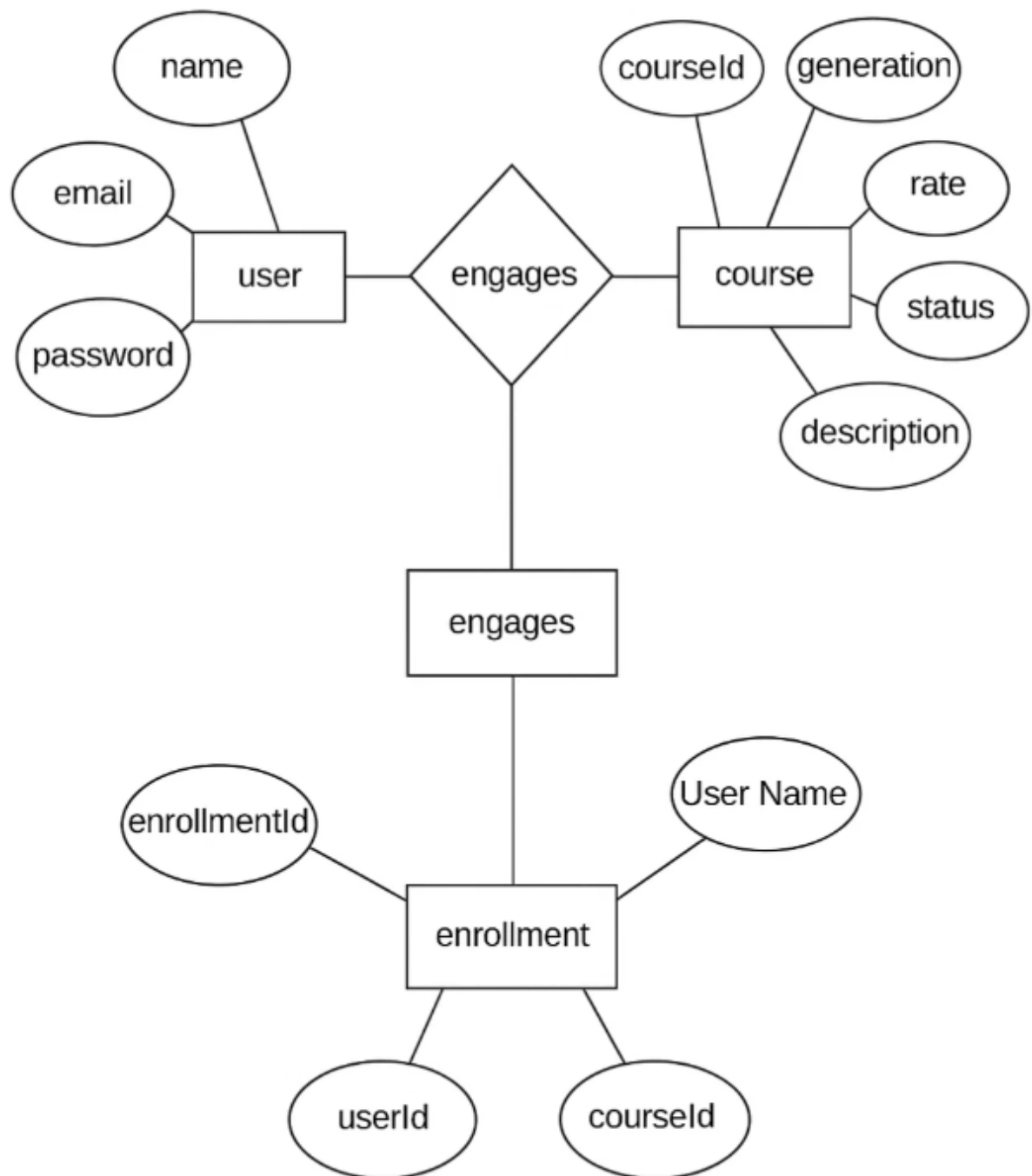
- **Frontend: React.js, Axios, Bootstrap, Material UI**
 - **Backend: Node.js, Express.js**
 - **Database: MongoDB, Mongoose**
 - **Others: JWT for authentication, Multer for file handling (if content upload is required), and Moment.js for time handling in progress tracking or deadlines**
-

Outcome

This architecture allows LearnHub to deliver:

- **Real-time, personalized learning experiences**
- **Multi-role support (learners, trainers, admins)**
- **Offline and mobile-friendly education**
- **Scalable infrastructure to expand across districts or states**
- **Reliable performance even in low-connectivity regions**

ER DIAGRAM



4. SETUP INSTRUCTIONS

- PREREQUISITE AND INSTALLATION:

Below are the key prerequisites and tools required for developing the LearnHub full-stack application using **Node.js, Express.js, MongoDB, and React.js** (MERN stack):

✓ Node.js and npm

Node.js is used to build the backend server logic for LearnHub. It allows JavaScript to be run server-side, while npm helps manage project dependencies.

-  **Download:** <https://nodejs.org/en/download/>
-  **Installation Guide:** <https://nodejs.org/en/download/package-manager/>
-  **Command:**

```
bash
```

```
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```

```
npm init
```

✓ Express.js

Express.js simplifies server-side logic and is used to manage routes, API calls, authentication, and learning content delivery.

-  **Install Express:**

```
bash
```

```
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```

```
npm install express
```

✓ MongoDB

MongoDB is used to store learner profiles, courses, quiz data, and progress tracking information in a flexible JSON-like format.

-  **Download:** <https://www.mongodb.com/try/download/community>
 -  **Install Guide:** <https://docs.mongodb.com/manual/installation/>
-

✓ Mongoose

Mongoose acts as a bridge between the Node.js backend and MongoDB, enabling schema definitions and simplifying data operations.

-  **Install Mongoose:**

bash

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```
npm install mongoose
```

✓ **React.js**

React powers the LearnHub frontend, offering reusable UI components for learners, trainers, and administrators.

-  **Installation Guide:** <https://reactjs.org/docs/create-a-new-react-app.html>
-  **Command:**

bash

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```
npx create-react-app learnhub-frontend
```

✓ **Ant Design (AntD)**

AntD is used to design elegant user interfaces for course dashboards, progress reports, and admin panels.

-  **AntD Docs:** <https://ant.design/docs/react/introduce>
-  **Command:**

bash

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```
npm install antd
```

✓ **Moment.js**

Used to format timestamps and track activity logs or course progress deadlines in human-readable formats.

-  **Moment.js:** <https://momentjs.com/>
-  **Command:**

bash

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npm install moment

✓ **HTML, CSS, JavaScript**

Basic web development skills are needed to customize layouts, styles, and interactivity.

✓ **Database Connectivity with Mongoose**

To perform CRUD operations for users, courses, modules, quizzes, and progress.

-  **Guide:** <https://www.section.io/engineering-education/nodejs-mongoosejs-mongodb/>
-

✓ **Version Control (Git)**

Use Git for source control and collaboration. Host your repository on GitHub.

-  **Download Git:** <https://git-scm.com/downloads>
-

✓ **Code Editor**

- Recommended: **Visual Studio Code**
 -  **Download:** <https://code.visualstudio.com/download>
-

• **PROJECT SETUP INSTRUCTIONS**

To run the existing LearnHub app from GitHub:

1. **Clone the Repository**

bash

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git clone https://github.com/your-username/learnhub.git

cd learnhub

2. Install Dependencies

bash

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cd frontend

npm install

cd ../backend

npm install

3. Start the Development Server

bash

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npm start

4. Access LearnHub

Open your browser and visit:

<http://localhost:3000>

• ROLES AND RESPONSIBILITIES

Learner:

1. **Register/Login** to the platform using email credentials.
2. Browse available courses/modules in their preferred language.
3. Enroll in courses and track progress via dashboard.
4. Take quizzes and receive certificates upon course completion.

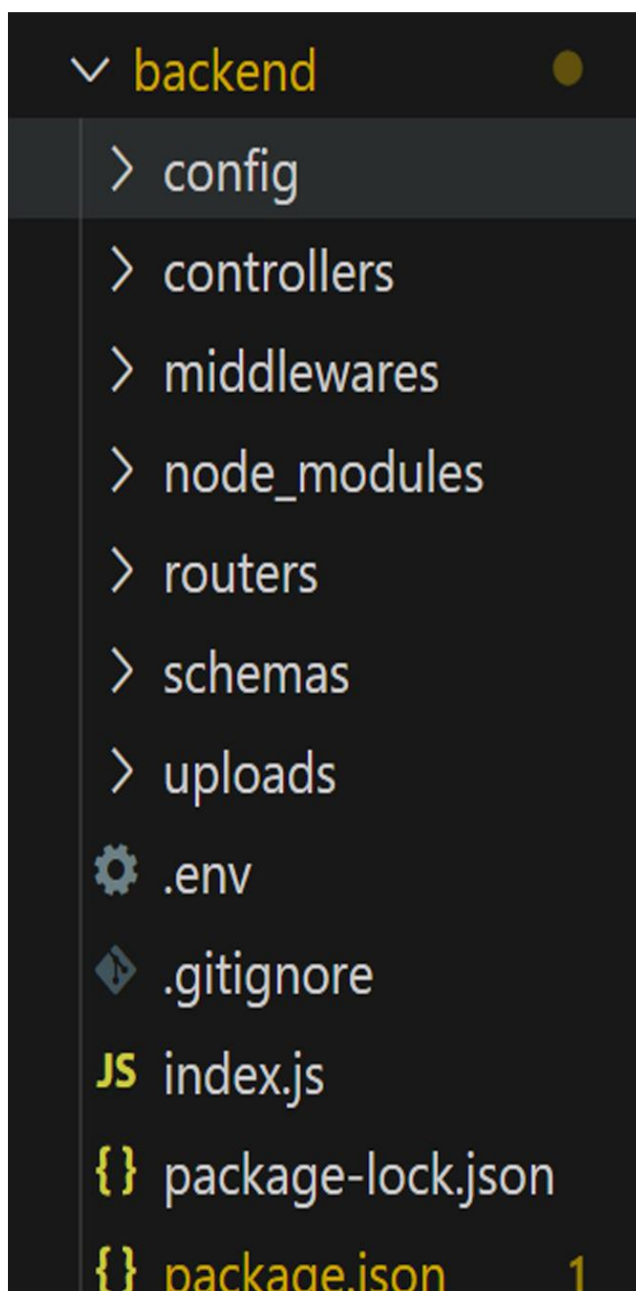
Trainer/Educator:

1. Upload new courses or lessons (video, text, quizzes).
2. Monitor learner progress and quiz scores.
3. Provide feedback or answer learner questions via message boards or forums.
4. Update or revise course content when necessary.

 Admin:

1. Approve educator/trainer accounts.
2. Manage course library and categories.
3. Monitor platform usage and analytics.
4. Ensure quality and compliance of learning material.
5. Enforce user policies, content guidelines, and data privacy.

5.FOLDER STRUCTURE



▼ frontend

> node_modules

> src

 .eslintrc.cjs

 .gitignore

 index.html

 package-lock.json

 package.json 1

 README.md

 vite.config.js

> node_modules

 package-lock.json

 package.json

The first image is of backend part which is showing all the files and folders that have been used in UI development

The second image is of frontend part which is showing all the files and folders that have been used in backend development

6. Project Flow

Milestone 1: Project Setup and Configuration

Folder Setup:

1. Create two main folders at the root level:

- /frontend – for the React-based user interface
 - /backend – for the Node.js + Express.js server
-

Installation of Required Tools and Libraries

Frontend Setup

Navigate into the /frontend folder and install the following packages:

bash

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npm install react bootstrap @mui/material axios moment antd
mdb-react-ui-kit react-bootstrap

Frontend Dependencies Used:

- react – For building user interfaces
- bootstrap – For responsive styling

- **@mui/material** – Material UI for React components
- **axios** – For making HTTP requests
- **moment** – For formatting date/time
- **antd** – Ant Design UI components
- **mdb-react-ui-kit** – Material Design for Bootstrap (React version)
- **react-bootstrap** – Bootstrap components as React components

The structure also includes `vite.config.js` as the project is using Vite as a build tool for faster development and bundling.

Backend Setup

Navigate into the `/backend` folder and install the following packages:

bash

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npm install express cors mongoose dotenv bcryptjs multer moment jsonwebtoken

npm install --save-dev nodemon

Backend Dependencies Used:

- **express** – Web server and API routing
- **cors** – Cross-Origin Resource Sharing setup
- **mongoose** – ODM for MongoDB
- **dotenv** – Load environment variables from `.env`
- **bcryptjs** – Password hashing for user authentication

- **multer** – For handling file uploads (e.g., profile pictures, documents)
- **moment** – Date and time formatting
- **jsonwebtoken** – For secure authentication (JWT)
- **nodemon** – Auto-restart development server on changes (dev dependency)

Backend Folder Structure Explanation:

- **config/** – Database configuration files
- **controllers/** – Logic for handling incoming requests
- **middlewares/** – Custom middleware (e.g., auth checks)
- **routers/** – API routing files
- **schemas/** – Mongoose schema models (User, Course, etc.)
- **uploads/** – For storing uploaded files (optional)
- **.env** – Environment variables
- **index.js** – Main server entry point

```
{
  "name": "frontend",
  "private": true,
  "version": "0.0.0",
  "type": "module",
  >Debug
  "scripts": {
    "dev": "vite",
    "build": "vite build",
    "preview": "vite preview",
    "lint": "eslint . --ext js,jsx --report-unused-disable-directives --max-warnings 0"
  },
  "dependencies": {
    "@emotion/react": "^11.11.1",
    "@emotion/styled": "^11.11.0",
    "@mui/icons-material": "^5.14.9",
    "@mui/material": "^5.14.9",
    "axios": "^1.5.0",
    "bootstrap": "^5.3.2",
    "html2canvas": "^1.4.1"
```



```

    "jspdf": "^2.5.1",
    "mdb-react-ui-kit": "^6.1.0",
    "mdb-ui-kit": "^6.4.0",
    "react": "^18.2.0",
    "react-bootstrap": "^2.8.0",
    "react-dom": "^18.2.0",
    "react-player": "^2.13.0",
    "react-router-dom": "^6.16.0"
  },
  "devDependencies": {
    "@types/react": "^18.2.15",
    "@types/react-dom": "^18.2.7",
    "@vitejs/plugin-react": "^4.0.3",
    "eslint": "^8.45.0",
    "eslint-plugin-react": "^7.32.2",
    "eslint-plugin-react-hooks": "^4.6.0",
    "eslint-plugin-react-refresh": "^0.4.3",
    "vite": "^4.4.5"
  }
}

```

7.RUNNIG THE APPLICATION

- **Run Backend (Express + Node)**

Go to the backend folder

```
cd backend
```

```
npm install # install dependencies
```

```
npm run dev
```

Backend API: Express runs on localhost:5000

- **Run Frontend (React)**

Go to the client folder:

```
cd client
```

```
npm install # install dependencies
```

```
npm start # runs app on http://localhost:3000
```

8. API DOCUMENTATION

The LearnHub backend exposes a set of RESTful APIs to interact with the application's core features such as user management, course access, enrollment, and admin control.

Endpoints include:

User Endpoints

- POST /api/users/register
Register a new user (student or instructor).
 - POST /api/users/login
Authenticate user and generate JWT token.
 - GET /api/users/profile
Retrieve user profile details (protected route).
 - PUT /api/users/profile
Update user profile information.
-

Course Endpoints

- GET /api/courses
Fetch all available courses.
 - GET /api/courses/:id
Fetch a single course by ID (includes description, instructor, etc.).
 - POST /api/courses
Add a new course (admin or instructor only).
 - PUT /api/courses/:id
Edit an existing course (admin or instructor only).
 - DELETE /api/courses/:id
Delete a course (admin only).
-

Enrollment & Learning Endpoints

- POST /api/enrollments
Enroll a user in a course.

- GET /api/enrollments/my
View enrolled courses for the logged-in user.
 - POST /api/progress/update
Update user's learning progress.
-

Admin Endpoints

- GET /api/admin/users
Get a list of all registered users (admin only).
- DELETE /api/admin/users/:id
Delete a specific user account (admin only).
- GET /api/admin/stats
Fetch platform analytics (total users, courses, enrollments, etc.).

9. Authentication

Authentication in LearnHub ensures that user data and learning resources are securely accessed based on defined roles and permissions.

Key Features:

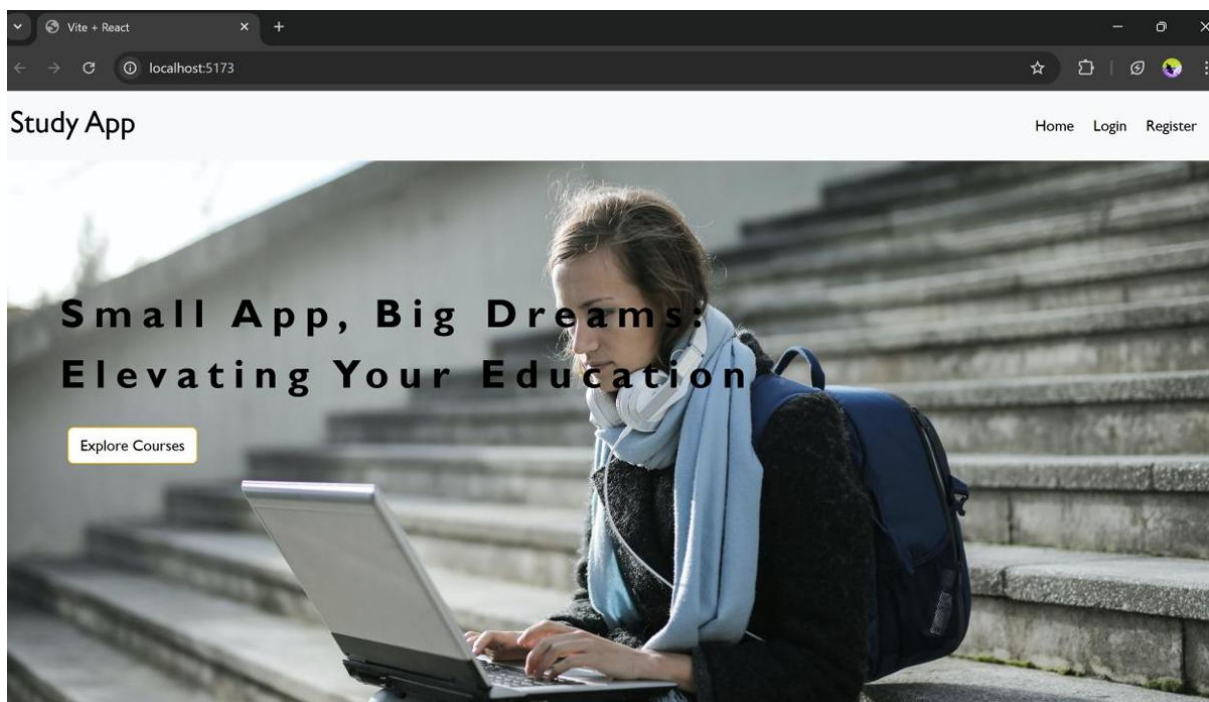
- **JWT-Based Authentication:**
LearnHub uses **JSON Web Tokens (JWT)** to securely authenticate users. Upon successful login or registration, a token is generated and stored in the HTTP headers for all subsequent requests.
- **Role-Based Access Control:**
Different types of users have different levels of access:
 - **Student:** Can browse and enroll in courses, track progress.
 - **Instructor:** Can create, update, and manage courses.
 - **Admin:** Has full control over users, courses, and platform analytics.
- **Protected Routes with Middleware:**
Custom authentication middleware ensures that only authorized users can access specific routes based on their roles. Unauthorized access is blocked with proper error handling.

10.USER INTERFACE

Testing the UI includes verifying the look and feel of each

page—landing, login, registration, and dashboards for different user types. Ensuring responsive design and usability across devices and screen sizes is also essential. **FINAL DEPLOYMENT:** Once testing is complete, the application can be deployed to a production server, making it accessible to end-users.

Landing Page:



Registration Page:

Study App

Home Login Register

Register

Full Name
rahul

Email Address
rahul@gmail.com

Password

Select User

Student
Teacher

Have an account? [Sign In](#)

Login Page:

Study App

Home Login Register

Sign In

Email Address
rahul

Password

SIGN IN

Have an account? [Sign Up](#)

CONCLUSION: LearnHub: Your Center for Skill Enhancement is a MERN stack e-learning platform with role-based access for learners, instructors, and admins. It features secure authentication, course management, and a responsive user interface. The project ensures scalability, modularity, and a seamless digital learning experience.